

CORE PURE / COMPLEX NUMBERS

De Moivre's Theorem

Worked solutions

18 questions · 180 marks

Questions

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1. A complex number z has modulus 1 and argument θ .

(a) Show that

$$z^n + \frac{1}{z^n} = 2 \cos n\theta \quad \text{and} \quad z^n - \frac{1}{z^n} = 2i \sin n\theta, \quad n \in \mathbb{Z}^+ \quad [3]$$

(b) Hence, by expanding $(z + \frac{1}{z})^3$ and $(z - \frac{1}{z})^3$, show that

$$\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \quad [4]$$

Solution

(a) Since $|z| = 1$ and $\arg z = \theta$,

$$z = \cos \theta + i \sin \theta = e^{i\theta}$$

So, by de Moivre's theorem,

$$z^n = e^{in\theta} = \cos n\theta + i \sin n\theta$$

Also,

$$\frac{1}{z} = e^{-i\theta}$$

and hence

$$\frac{1}{z^n} = e^{-in\theta} = \cos n\theta - i \sin n\theta$$

Now add:

$$z^n + \frac{1}{z^n} = (\cos n\theta + i \sin n\theta) + (\cos n\theta - i \sin n\theta) = 2 \cos n\theta$$

Now subtract:

$$z^n - \frac{1}{z^n} = (\cos n\theta + i \sin n\theta) - (\cos n\theta - i \sin n\theta) = 2i \sin n\theta$$

Therefore

$$z^n + \frac{1}{z^n} = 2 \cos n\theta \quad \text{and} \quad z^n - \frac{1}{z^n} = 2i \sin n\theta$$

(b) From part (a), with $n = 1$,

$$z + \frac{1}{z} = 2 \cos \theta$$

Cubing both sides gives

$$\left(z + \frac{1}{z}\right)^3 = (2 \cos \theta)^3 = 8 \cos^3 \theta$$

Expand the left-hand side:

$$\begin{aligned} \left(z + \frac{1}{z}\right)^3 &= z^3 + 3z + \frac{3}{z} + \frac{1}{z^3} \\ &= \left(z^3 + \frac{1}{z^3}\right) + 3\left(z + \frac{1}{z}\right) \end{aligned}$$

Using part (a) again,

$$8 \cos^3 \theta = 2 \cos 3\theta + 6 \cos \theta$$

So

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$

Now use part (a) with

$$z - \frac{1}{z} = 2i \sin \theta$$

Cubing both sides,

$$\left(z - \frac{1}{z}\right)^3 = (2i \sin \theta)^3 = -8i \sin^3 \theta$$

Expand:

$$\begin{aligned}\left(z - \frac{1}{z}\right)^3 &= z^3 - 3z + \frac{3}{z} - \frac{1}{z^3} \\ &= \left(z^3 - \frac{1}{z^3}\right) - 3\left(z - \frac{1}{z}\right)\end{aligned}$$

Using part (a),

$$-8i \sin^3 \theta = 2i \sin 3\theta - 6i \sin \theta$$

Hence

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$$

Let $t = \tan \theta$. Then

$$\begin{aligned}\tan 3\theta &= \frac{\sin 3\theta}{\cos 3\theta} \\ &= \frac{\frac{\sin 3\theta}{\cos^3 \theta}}{\frac{\cos 3\theta}{\cos^3 \theta}} \\ &= \frac{\frac{3 \sin \theta - 4 \sin^3 \theta}{\cos^3 \theta}}{\frac{4 \cos^3 \theta - 3 \cos \theta}{\cos^3 \theta}} \\ &= \frac{3 \tan \theta \sec^2 \theta - 4 \tan^3 \theta}{4 - 3 \sec^2 \theta} \\ &= \frac{3t(1+t^2) - 4t^3}{4 - 3(1+t^2)} \\ &= \frac{3t - t^3}{1 - 3t^2}\end{aligned}$$

Therefore

$$\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

2. (a) Using de Moivre's theorem, show that

$$\sin^4 \theta \cos^2 \theta \equiv \frac{1}{32}(\cos 6\theta - 2 \cos 4\theta - \cos 2\theta + 2) \quad [5]$$

(b) By using the substitution $\alpha = (\frac{\pi}{2} - \theta)$, or otherwise, find a similar identity for $\sin^2 \theta \cos^4 \theta$. [3]

(c) Given that $0 < a < \frac{\pi}{2}$ and

$$\int_0^a (\sin^4 \theta \cos^2 \theta + 2 \sin^2 \theta \cos^4 \theta) \, d\theta = \frac{\pi}{32} - \frac{\sqrt{3}}{64}$$

find the exact value of a .

[5]

Solution

(a) Let

$$z = \cos \theta + i \sin \theta$$

By de Moivre's theorem,

$$z^n = \cos n\theta + i \sin n\theta, \quad z^{-n} = \cos n\theta - i \sin n\theta$$

so

$$z^n + z^{-n} = 2 \cos n\theta, \quad z^n - z^{-n} = 2i \sin n\theta$$

In particular, $z + z^{-1} = 2 \cos \theta$ and $z - z^{-1} = 2i \sin \theta$, so

$$\begin{aligned} (z - z^{-1})^4 (z + z^{-1})^2 &= (2i \sin \theta)^4 (2 \cos \theta)^2 \\ &= 16i^4 \sin^4 \theta \cdot 4 \cos^2 \theta \\ &= 64 \sin^4 \theta \cos^2 \theta \end{aligned}$$

Now expand the left-hand side. Since $(z - z^{-1})(z + z^{-1}) = z^2 - z^{-2}$,

$$\begin{aligned} (z - z^{-1})^4 (z + z^{-1})^2 &= (z - z^{-1})^2 (z^2 - z^{-2})^2 \\ &= (z^2 - 2 + z^{-2})(z^4 - 2 + z^{-4}) \\ &= z^6 - 2z^2 + z^{-2} - 2z^4 + 4 - 2z^{-4} + z^2 - 2z^{-2} + z^{-6} \\ &= (z^6 + z^{-6}) - 2(z^4 + z^{-4}) - (z^2 + z^{-2}) + 4 \\ &= 2 \cos 6\theta - 4 \cos 4\theta - 2 \cos 2\theta + 4 \end{aligned}$$

Equating the two expressions,

$$64 \sin^4 \theta \cos^2 \theta = 2 \cos 6\theta - 4 \cos 4\theta - 2 \cos 2\theta + 4$$

so

$$\sin^4 \theta \cos^2 \theta \equiv \frac{1}{32} (\cos 6\theta - 2 \cos 4\theta - \cos 2\theta + 2)$$

Alternative method (using double-angle identities):

Let

$$c = \cos 2\theta$$

Using the double-angle identities,

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2} = \frac{1 - c}{2}, \quad \cos^2 \theta = \frac{1 + \cos 2\theta}{2} = \frac{1 + c}{2}$$

So

$$\begin{aligned}\sin^4 \theta \cos^2 \theta &= \left(\frac{1-c}{2}\right)^2 \left(\frac{1+c}{2}\right) \\&= \frac{1}{8}(1-2c+c^2)(1+c) \\&= \frac{1}{8}(1+c-2c-2c^2+c^2+c^3) \\&= \frac{1}{8}(1-c-c^2+c^3)\end{aligned}$$

Now use de Moivre's theorem.

From

$$(\cos 2\theta + i \sin 2\theta)^2 = \cos 4\theta + i \sin 4\theta$$

taking real parts gives

$$\cos 4\theta = 2 \cos^2 2\theta - 1 = 2c^2 - 1$$

so

$$c^2 = \frac{1 + \cos 4\theta}{2}$$

Also, from

$$(\cos 2\theta + i \sin 2\theta)^3 = \cos 6\theta + i \sin 6\theta$$

taking real parts gives

$$\cos 6\theta = 4 \cos^3 2\theta - 3 \cos 2\theta = 4c^3 - 3c$$

so

$$c^3 = \frac{\cos 6\theta + 3c}{4}$$

Substituting these into $\frac{1}{8}(1-c-c^2+c^3)$,

$$\begin{aligned}\sin^4 \theta \cos^2 \theta &= \frac{1}{8} \left(1 - c - \frac{1 + \cos 4\theta}{2} + \frac{\cos 6\theta + 3c}{4}\right) \\&= \frac{1}{8} \left(\frac{4 - 4c - 2 - 2 \cos 4\theta + \cos 6\theta + 3c}{4}\right) \\&= \frac{1}{32} (2 - c - 2 \cos 4\theta + \cos 6\theta)\end{aligned}$$

Since $c = \cos 2\theta$,

$$\sin^4 \theta \cos^2 \theta \equiv \frac{1}{32} (\cos 6\theta - 2 \cos 4\theta - \cos 2\theta + 2)$$

Hence the required identity is shown.

(b) Let

$$\alpha = \frac{\pi}{2} - \theta$$

Then

$$\sin \theta = \cos \alpha, \quad \cos \theta = \sin \alpha$$

so

$$\sin^2 \theta \cos^4 \theta = \cos^2 \alpha \sin^4 \alpha = \sin^4 \alpha \cos^2 \alpha$$

Using part (a),

$$\sin^4 \alpha \cos^2 \alpha = \frac{1}{32} (\cos 6\alpha - 2 \cos 4\alpha - \cos 2\alpha + 2)$$

Now

$$2\alpha = \pi - 2\theta, \quad 4\alpha = 2\pi - 4\theta, \quad 6\alpha = 3\pi - 6\theta$$

so

$$\cos 2\alpha = -\cos 2\theta, \quad \cos 4\alpha = \cos 4\theta, \quad \cos 6\alpha = -\cos 6\theta$$

Therefore

$$\sin^2 \theta \cos^4 \theta = \frac{1}{32} (-\cos 6\theta - 2 \cos 4\theta + \cos 2\theta + 2)$$

So a similar identity is

$$\sin^2 \theta \cos^4 \theta \equiv \frac{1}{32} (-\cos 6\theta - 2 \cos 4\theta + \cos 2\theta + 2)$$

(c) Let

$$I(a) = \int_0^a (\sin^4 \theta \cos^2 \theta + 2 \sin^2 \theta \cos^4 \theta) \, d\theta$$

Using parts (a) and (b),

$$\begin{aligned} \sin^4 \theta \cos^2 \theta + 2 \sin^2 \theta \cos^4 \theta &= \frac{1}{32} (\cos 6\theta - 2 \cos 4\theta - \cos 2\theta + 2) \\ &\quad + \frac{2}{32} (-\cos 6\theta - 2 \cos 4\theta + \cos 2\theta + 2) \\ &= \frac{1}{32} (-\cos 6\theta - 6 \cos 4\theta + \cos 2\theta + 6) \end{aligned}$$

Hence

$$\begin{aligned} I(a) &= \frac{1}{32} \int_0^a (-\cos 6\theta - 6 \cos 4\theta + \cos 2\theta + 6) \, d\theta \\ &= \frac{1}{32} \left[-\frac{1}{6} \sin 6\theta - \frac{3}{2} \sin 4\theta + \frac{1}{2} \sin 2\theta + 6\theta \right]_0^a \\ &= \frac{1}{32} \left(6a + \frac{1}{2} \sin 2a - \frac{3}{2} \sin 4a - \frac{1}{6} \sin 6a \right) \end{aligned}$$

We are given that

$$I(a) = \frac{\pi}{32} - \frac{\sqrt{3}}{64}$$

Try $a = \frac{\pi}{6}$. Then

$$6a = \pi, \quad \sin 2a = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}, \quad \sin 4a = \sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2}, \quad \sin 6a = \sin \pi = 0$$

So

$$\begin{aligned} I\left(\frac{\pi}{6}\right) &= \frac{1}{32} \left(\pi + \frac{1}{2} \cdot \frac{\sqrt{3}}{2} - \frac{3}{2} \cdot \frac{\sqrt{3}}{2} - 0 \right) \\ &= \frac{1}{32} \left(\pi - \frac{\sqrt{3}}{2} \right) \\ &= \frac{\pi}{32} - \frac{\sqrt{3}}{64} \end{aligned}$$

So $a = \frac{\pi}{6}$ satisfies the equation.

To show this is the only solution, differentiate $I(a)$ using the Fundamental Theorem of Calculus:

$$I'(a) = \sin^4 a \cos^2 a + 2 \sin^2 a \cos^4 a$$

and factorise:

$$I'(a) = \sin^2 a \cos^2 a (\sin^2 a + 2 \cos^2 a)$$

For $0 < a < \frac{\pi}{2}$, we have $\sin a > 0$ and $\cos a > 0$, so

$$I'(a) > 0$$

Thus $I(a)$ is strictly increasing on $(0, \frac{\pi}{2})$, so the solution is unique.

Therefore

$$a = \frac{\pi}{6}$$

that is, $a = 30^\circ$.

3. Use de Moivre's theorem to find the constants A , B and C in the identity

$$\sin \theta \cos^4 \theta = A \sin \theta + B \sin 3\theta + C \sin 5\theta \quad [4]$$

Solution

Let

$$z = \cos \theta + i \sin \theta$$

Then $z^{-1} = \cos \theta - i \sin \theta$, so

$$\cos \theta = \frac{z + z^{-1}}{2}, \quad \sin \theta = \frac{z - z^{-1}}{2i}$$

Also, by de Moivre's theorem,

$$z^n = \cos n\theta + i \sin n\theta$$

which gives

$$\sin n\theta = \frac{z^n - z^{-n}}{2i}$$

Now write $\sin \theta \cos^4 \theta$ in terms of z :

$$\begin{aligned} \sin \theta \cos^4 \theta &= \frac{z - z^{-1}}{2i} \left(\frac{z + z^{-1}}{2} \right)^4 \\ &= \frac{1}{32i} (z - z^{-1})(z + z^{-1})^4 \end{aligned}$$

Expand $(z + z^{-1})^4$:

$$(z + z^{-1})^4 = z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4}$$

So

$$\begin{aligned} \sin \theta \cos^4 \theta &= \frac{1}{32i} (z - z^{-1})(z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4}) \\ &= \frac{1}{32i} (z^5 + 4z^3 + 6z + 4z^{-1} + z^{-3} \\ &\quad - z^3 - 4z - 6z^{-1} - 4z^{-3} - z^{-5}) \\ &= \frac{1}{32i} (z^5 + 3z^3 + 2z - 2z^{-1} - 3z^{-3} - z^{-5}) \\ &= \frac{1}{32i} [(z^5 - z^{-5}) + 3(z^3 - z^{-3}) + 2(z - z^{-1})] \end{aligned}$$

Now use

$$\frac{z^n - z^{-n}}{2i} = \sin n\theta$$

Hence

$$\begin{aligned} \sin \theta \cos^4 \theta &= \frac{1}{16} \sin 5\theta + \frac{3}{16} \sin 3\theta + \frac{2}{16} \sin \theta \\ &= \frac{1}{8} \sin \theta + \frac{3}{16} \sin 3\theta + \frac{1}{16} \sin 5\theta \end{aligned}$$

Comparing with

$$\sin \theta \cos^4 \theta = A \sin \theta + B \sin 3\theta + C \sin 5\theta$$

gives

$$A = \frac{1}{8}, \quad B = \frac{3}{16}, \quad C = \frac{1}{16}$$

4. (i) Starting with the result

$$e^{i\theta} = \cos \theta + i \sin \theta$$

show that

(A)

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta \quad [2]$$

(B)

$$\sin \theta = \frac{1}{2i} (e^{i\theta} - e^{-i\theta}) \quad [2]$$

(ii) Using the result in part (i)(A), obtain the values of the constants a, b, c, d and e in the identity

$$\cos 8\theta \equiv a \sin^8 \theta + b \sin^6 \theta + c \sin^4 \theta + d \sin^2 \theta + e \quad [6]$$

(iii) Using the result in part (i)(B), obtain the values of the constants P, Q, R, S and T in the identity

$$\sin^8 \theta \equiv P \cos 8\theta + Q \cos 6\theta + R \cos 4\theta + S \cos 2\theta + T \quad [6]$$

(iv) Hence show that

$$\sin \frac{\pi}{8} = \left(\frac{17 - 12\sqrt{2}}{64} \right)^{\frac{1}{8}} \quad [3]$$

Solution

(i) (A) Starting from

$$e^{i\theta} = \cos \theta + i \sin \theta$$

raise both sides to the power n :

$$(\cos \theta + i \sin \theta)^n = (e^{i\theta})^n = e^{in\theta}$$

Now apply Euler's formula again, with θ replaced by $n\theta$:

$$e^{in\theta} = \cos n\theta + i \sin n\theta$$

Hence

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

(B) Replace θ by $-\theta$ in Euler's formula:

$$e^{-i\theta} = \cos(-\theta) + i \sin(-\theta) = \cos \theta - i \sin \theta$$

Subtract this from

$$e^{i\theta} = \cos \theta + i \sin \theta$$

to get

$$e^{i\theta} - e^{-i\theta} = 2i \sin \theta$$

Therefore

$$\sin \theta = \frac{1}{2i} (e^{i\theta} - e^{-i\theta})$$

(ii) Let $C = \cos \theta$ and $S = \sin \theta$. From part (i)(A),

$$(C + iS)^8 = \cos 8\theta + i \sin 8\theta$$

Expand $(C + iS)^8$:

$$\begin{aligned}(C + iS)^8 &= C^8 + 8iC^7S + 28C^6(iS)^2 + 56C^5(iS)^3 + 70C^4(iS)^4 \\ &\quad + 56C^3(iS)^5 + 28C^2(iS)^6 + 8C(iS)^7 + (iS)^8 \\ &= C^8 + 8iC^7S - 28C^6S^2 - 56iC^5S^3 + 70C^4S^4 \\ &\quad + 56iC^3S^5 - 28C^2S^6 - 8iCS^7 + S^8\end{aligned}$$

Taking real parts,

$$\cos 8\theta = C^8 - 28C^6S^2 + 70C^4S^4 - 28C^2S^6 + S^8$$

Since $C^2 = 1 - S^2$, let $x = S^2$. Then

$$\begin{aligned}\cos 8\theta &= (1 - x)^4 - 28(1 - x)^3x + 70(1 - x)^2x^2 - 28(1 - x)x^3 + x^4 \\ &= (1 - 4x + 6x^2 - 4x^3 + x^4) \\ &\quad - 28(x - 3x^2 + 3x^3 - x^4) \\ &\quad + 70(x^2 - 2x^3 + x^4) - 28(x^3 - x^4) + x^4 \\ &= 1 - 32x + 160x^2 - 256x^3 + 128x^4\end{aligned}$$

Substitute back $x = \sin^2 \theta$:

$$\cos 8\theta \equiv 128 \sin^8 \theta - 256 \sin^6 \theta + 160 \sin^4 \theta - 32 \sin^2 \theta + 1$$

Hence

$$a = 128, \quad b = -256, \quad c = 160, \quad d = -32, \quad e = 1$$

(iii) From part (i)(B),

$$\sin \theta = \frac{1}{2i} (e^{i\theta} - e^{-i\theta})$$

Therefore

$$\sin^8 \theta = \left(\frac{1}{2i} \right)^8 (e^{i\theta} - e^{-i\theta})^8 = \frac{1}{256} (e^{i\theta} - e^{-i\theta})^8$$

Now expand:

$$\begin{aligned}\sin^8 \theta &= \frac{1}{256} \left(e^{8i\theta} - 8e^{6i\theta} + 28e^{4i\theta} - 56e^{2i\theta} + 70 \right. \\ &\quad \left. - 56e^{-2i\theta} + 28e^{-4i\theta} - 8e^{-6i\theta} + e^{-8i\theta} \right)\end{aligned}$$

Use

$$e^{ki\theta} + e^{-ki\theta} = 2 \cos k\theta$$

so that

$$\begin{aligned}\sin^8 \theta &= \frac{1}{256} \left((e^{8i\theta} + e^{-8i\theta}) - 8(e^{6i\theta} + e^{-6i\theta}) \right. \\ &\quad \left. + 28(e^{4i\theta} + e^{-4i\theta}) - 56(e^{2i\theta} + e^{-2i\theta}) + 70 \right) \\ &= \frac{1}{256} \left(2 \cos 8\theta - 16 \cos 6\theta + 56 \cos 4\theta - 112 \cos 2\theta + 70 \right) \\ &= \frac{1}{128} \cos 8\theta - \frac{1}{16} \cos 6\theta + \frac{7}{32} \cos 4\theta - \frac{7}{16} \cos 2\theta + \frac{35}{128}\end{aligned}$$

So

$$P = \frac{1}{128}, \quad Q = -\frac{1}{16}, \quad R = \frac{7}{32}, \quad S = -\frac{7}{16}, \quad T = \frac{35}{128}$$

Equivalently,

$$\sin^8 \theta \equiv \frac{\cos 8\theta - 8 \cos 6\theta + 28 \cos 4\theta - 56 \cos 2\theta + 35}{128}$$

(iv) Using the identity from part (iii) with $\theta = \frac{\pi}{8}$,

$$\sin^8 \frac{\pi}{8} = \frac{\cos \pi - 8 \cos \frac{3\pi}{4} + 28 \cos \frac{\pi}{2} - 56 \cos \frac{\pi}{4} + 35}{128}$$

Now

$$\cos \pi = -1, \quad \cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2}, \quad \cos \frac{\pi}{2} = 0, \quad \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

so

$$\begin{aligned} \sin^8 \frac{\pi}{8} &= \frac{-1 - 8 \left(-\frac{\sqrt{2}}{2}\right) + 28(0) - 56 \left(\frac{\sqrt{2}}{2}\right) + 35}{128} \\ &= \frac{-1 + 4\sqrt{2} - 28\sqrt{2} + 35}{128} \\ &= \frac{34 - 24\sqrt{2}}{128} \\ &= \frac{17 - 12\sqrt{2}}{64} \end{aligned}$$

Since $0 < \frac{\pi}{8} < \frac{\pi}{2}$, $\sin \frac{\pi}{8} > 0$. Hence

$$\sin \frac{\pi}{8} = \left(\frac{17 - 12\sqrt{2}}{64} \right)^{\frac{1}{8}}$$

5. (a) Use de Moivre's theorem to show that $\cos 6\theta \equiv (1 - 2\sin^2 \theta)(16\sin^4 \theta - 16\sin^2 \theta + 1)$ [5]

(b) By solving the equation $\cos 6\theta = 0$, deduce that $\sin^2 \left(\frac{\pi}{12} \right) = \frac{2-\sqrt{3}}{4}$. [4]

(c) Hence, or otherwise, write down the exact values of $\sin^2 \left(\frac{5\pi}{12} \right)$, $\sin^2 \left(\frac{7\pi}{12} \right)$ and $\sin^2 \left(\frac{11\pi}{12} \right)$. [3]

Solution

(a) Using de Moivre's theorem,

$$(\cos \theta + i \sin \theta)^6 = \cos 6\theta + i \sin 6\theta$$

Expand the left-hand side by the binomial theorem:

$$\begin{aligned} (\cos \theta + i \sin \theta)^6 &= \cos^6 \theta + 6 \cos^5 \theta (i \sin \theta) + 15 \cos^4 \theta (i \sin \theta)^2 \\ &\quad + 20 \cos^3 \theta (i \sin \theta)^3 + 15 \cos^2 \theta (i \sin \theta)^4 \\ &\quad + 6 \cos \theta (i \sin \theta)^5 + (i \sin \theta)^6 \\ &= \cos^6 \theta + 6i \cos^5 \theta \sin \theta - 15 \cos^4 \theta \sin^2 \theta \\ &\quad - 20i \cos^3 \theta \sin^3 \theta + 15 \cos^2 \theta \sin^4 \theta \\ &\quad + 6i \cos \theta \sin^5 \theta - \sin^6 \theta \end{aligned}$$

Taking real parts gives

$$\cos 6\theta = \cos^6 \theta - 15 \cos^4 \theta \sin^2 \theta + 15 \cos^2 \theta \sin^4 \theta - \sin^6 \theta$$

Now let

$$u = \sin^2 \theta \quad \text{so} \quad \cos^2 \theta = 1 - u$$

Then

$$\begin{aligned} \cos 6\theta &= (1 - u)^3 - 15(1 - u)^2 u + 15(1 - u)u^2 - u^3 \\ &= (1 - 3u + 3u^2 - u^3) - 15(u - 2u^2 + u^3) + 15(u^2 - u^3) - u^3 \\ &= 1 - 3u + 3u^2 - u^3 - 15u + 30u^2 - 15u^3 + 15u^2 - 15u^3 - u^3 \\ &= 1 - 18u + 48u^2 - 32u^3 \end{aligned}$$

Now factorise:

$$1 - 18u + 48u^2 - 32u^3 = (1 - 2u)(16u^2 - 16u + 1)$$

Substituting back $u = \sin^2 \theta$,

$$\cos 6\theta = (1 - 2\sin^2 \theta)(16\sin^4 \theta - 16\sin^2 \theta + 1)$$

Hence

$$\cos 6\theta \equiv (1 - 2\sin^2 \theta)(16\sin^4 \theta - 16\sin^2 \theta + 1)$$

(b) Let

$$u = \sin^2 \theta$$

From part (a), if $\cos 6\theta = 0$, then

$$(1 - 2u)(16u^2 - 16u + 1) = 0$$

So either

$$1 - 2u = 0 \quad \Rightarrow \quad u = \frac{1}{2}$$

or

$$16u^2 - 16u + 1 = 0$$

$$\begin{aligned} u &= \frac{16 \pm \sqrt{(-16)^2 - 4(16)(1)}}{2 \cdot 16} \\ &= \frac{16 \pm \sqrt{256 - 64}}{32} \\ &= \frac{16 \pm \sqrt{192}}{32} \\ &= \frac{16 \pm 8\sqrt{3}}{32} \\ &= \frac{2 \pm \sqrt{3}}{4} \end{aligned}$$

Now

$$\cos\left(6 \cdot \frac{\pi}{12}\right) = \cos \frac{\pi}{2} = 0$$

so $\sin^2\left(\frac{\pi}{12}\right)$ must be one of

$$\frac{1}{2}, \quad \frac{2 + \sqrt{3}}{4}, \quad \frac{2 - \sqrt{3}}{4}$$

But

$$0 < \frac{\pi}{12} < \frac{\pi}{4}$$

so

$$\sin^2\left(\frac{\pi}{12}\right) < \sin^2\left(\frac{\pi}{4}\right) = \frac{1}{2}$$

Therefore the correct value is

$$\sin^2\left(\frac{\pi}{12}\right) = \frac{2 - \sqrt{3}}{4}$$

(c) Using part (b),

$$\begin{aligned} \sin^2\left(\frac{5\pi}{12}\right) &= \cos^2\left(\frac{\pi}{12}\right) \\ &= 1 - \sin^2\left(\frac{\pi}{12}\right) \\ &= 1 - \frac{2 - \sqrt{3}}{4} \\ &= \frac{2 + \sqrt{3}}{4} \end{aligned}$$

Also,

$$\sin^2\left(\frac{7\pi}{12}\right) = \sin^2\left(\pi - \frac{5\pi}{12}\right) = \sin^2\left(\frac{5\pi}{12}\right) = \frac{2 + \sqrt{3}}{4}$$

and

$$\sin^2\left(\frac{11\pi}{12}\right) = \sin^2\left(\pi - \frac{\pi}{12}\right) = \sin^2\left(\frac{\pi}{12}\right) = \frac{2 - \sqrt{3}}{4}$$

Hence

$$\sin^2\left(\frac{5\pi}{12}\right) = \frac{2 + \sqrt{3}}{4}, \quad \sin^2\left(\frac{7\pi}{12}\right) = \frac{2 + \sqrt{3}}{4}, \quad \sin^2\left(\frac{11\pi}{12}\right) = \frac{2 - \sqrt{3}}{4}$$

6. Let $t = \tan \theta$. By writing

$$\cos \theta + i \sin \theta = \cos \theta (1 + it)$$

and expanding $(1 + it)^6$, use de Moivre's theorem to show that

$$\tan 6\theta = \frac{6 \tan \theta - 20 \tan^3 \theta + 6 \tan^5 \theta}{1 - 15 \tan^2 \theta + 15 \tan^4 \theta - \tan^6 \theta} \quad [7]$$

Solution

Let

$$z = \cos \theta + i \sin \theta$$

and let

$$t = \tan \theta = \frac{\sin \theta}{\cos \theta}$$

Then $\sin \theta = t \cos \theta$, so

$$z = \cos \theta + i \sin \theta = \cos \theta + i(t \cos \theta) = \cos \theta (1 + it)$$

Now apply de Moivre's theorem:

$$z^6 = \cos 6\theta + i \sin 6\theta$$

But also, from $z = \cos \theta (1 + it)$,

$$z^6 = \cos^6 \theta (1 + it)^6$$

So we expand $(1 + it)^6$ using the binomial theorem:

$$(1 + it)^6 = 1 + 6(it) + 15(it)^2 + 20(it)^3 + 15(it)^4 + 6(it)^5 + (it)^6$$

Now simplify the powers of i :

$$i^2 = -1, \quad i^3 = -i, \quad i^4 = 1, \quad i^5 = i, \quad i^6 = -1$$

Hence

$$\begin{aligned} (1 + it)^6 &= 1 + 6it + 15(-1)t^2 + 20(-i)t^3 + 15(1)t^4 + 6(i)t^5 + (-1)t^6 \\ &= 1 + 6it - 15t^2 - 20it^3 + 15t^4 + 6it^5 - t^6 \\ &= (1 - 15t^2 + 15t^4 - t^6) + i(6t - 20t^3 + 6t^5) \end{aligned}$$

Therefore

$$\cos 6\theta + i \sin 6\theta = \cos^6 \theta \left[(1 - 15t^2 + 15t^4 - t^6) + i(6t - 20t^3 + 6t^5) \right]$$

Equating real and imaginary parts gives

$$\cos 6\theta = \cos^6 \theta (1 - 15t^2 + 15t^4 - t^6)$$

and

$$\sin 6\theta = \cos^6 \theta (6t - 20t^3 + 6t^5)$$

So

$$\begin{aligned} \tan 6\theta &= \frac{\sin 6\theta}{\cos 6\theta} \\ &= \frac{\cos^6 \theta (6t - 20t^3 + 6t^5)}{\cos^6 \theta (1 - 15t^2 + 15t^4 - t^6)} \\ &= \frac{6t - 20t^3 + 6t^5}{1 - 15t^2 + 15t^4 - t^6} \end{aligned}$$

Since $t = \tan \theta$, this becomes

$$\tan 6\theta = \frac{6 \tan \theta - 20 \tan^3 \theta + 6 \tan^5 \theta}{1 - 15 \tan^2 \theta + 15 \tan^4 \theta - \tan^6 \theta}$$

which is the required result.

7. (a) Use de Moivre's theorem to show that

$$\sin 6x = \sin 2x (a \cos^4 x + b \cos^2 x + c)$$

where a , b and c are integers to be determined.

[4]

(b) Hence solve, for $0 < \theta < \frac{\pi}{2}$,

$$\sin 6\theta = \sin 2\theta \cos 2\theta$$

giving your answers to 3 decimal places.

[4]

Solution

(a) Let $C = \cos x$ and $S = \sin x$.

By de Moivre's theorem,

$$(\cos x + i \sin x)^6 = \cos 6x + i \sin 6x$$

So

$$\begin{aligned} (C + iS)^6 &= C^6 + 6iC^5S + 15C^4(iS)^2 + 20C^3(iS)^3 + 15C^2(iS)^4 + 6C(iS)^5 + (iS)^6 \\ &= C^6 + 6iC^5S - 15C^4S^2 - 20iC^3S^3 + 15C^2S^4 + 6iCS^5 - S^6 \end{aligned}$$

Taking imaginary parts gives

$$\begin{aligned} \sin 6x &= 6C^5S - 20C^3S^3 + 6CS^5 \\ &= 2CS(3C^4 - 10C^2S^2 + 3S^4) \end{aligned}$$

Now simplify the bracket:

$$\begin{aligned} 3C^4 - 10C^2S^2 + 3S^4 &= 3(C^4 + S^4) - 10C^2S^2 \\ &= 3((C^2 + S^2)^2 - 2C^2S^2) - 10C^2S^2 \\ &= 3(1 - 2C^2S^2) - 10C^2S^2 \\ &= 3 - 16C^2S^2 \\ &= 3 - 16C^2(1 - C^2) \\ &= 16C^4 - 16C^2 + 3 \end{aligned}$$

Also, $\sin 2x = 2 \sin x \cos x = 2CS$, so

$$\sin 6x = \sin 2x (16 \cos^4 x - 16 \cos^2 x + 3)$$

Hence

$$a = 16, \quad b = -16, \quad c = 3$$

(b) Using part (a),

$$\sin 6\theta = \sin 2\theta (16 \cos^4 \theta - 16 \cos^2 \theta + 3)$$

So the equation

$$\sin 6\theta = \sin 2\theta \cos 2\theta$$

becomes

$$\sin 2\theta (16 \cos^4 \theta - 16 \cos^2 \theta + 3) = \sin 2\theta \cos 2\theta$$

Since $0 < \theta < \frac{\pi}{2}$, we have $0 < 2\theta < \pi$, so $\sin 2\theta \neq 0$.

Therefore we can divide by $\sin 2\theta$:

$$16 \cos^4 \theta - 16 \cos^2 \theta + 3 = \cos 2\theta$$

Let

$$u = \cos^2 \theta$$

Then $\cos 2\theta = 2u - 1$, so

$$16u^2 - 16u + 3 = 2u - 1$$

$$16u^2 - 18u + 4 = 0$$

Solving,

$$\begin{aligned} u &= \frac{18 \pm \sqrt{18^2 - 4(16)(4)}}{2(16)} \\ &= \frac{18 \pm \sqrt{324 - 256}}{32} \\ &= \frac{18 \pm \sqrt{68}}{32} \\ &= \frac{9 \pm \sqrt{17}}{16} \end{aligned}$$

Hence

$$\begin{aligned} \cos 2\theta &= 2u - 1 \\ &= 2 \left(\frac{9 \pm \sqrt{17}}{16} \right) - 1 \\ &= \frac{1 \pm \sqrt{17}}{8} \end{aligned}$$

So

$$2\theta = \arccos \left(\frac{1 \pm \sqrt{17}}{8} \right)$$

and therefore

$$\theta = \frac{1}{2} \arccos \left(\frac{1 \pm \sqrt{17}}{8} \right)$$

This gives

$$\theta \approx 0.438, 0.986$$

So the solutions are

$$\theta = 0.438 \text{ rad}, \quad 0.986 \text{ rad}$$

to 3 decimal places.

8. (a) By considering the binomial expansion of $(z - z^{-1})^6$, where $z = \cos \theta + i \sin \theta$, use de Moivre's theorem to show that

$$\sin^6 \theta = \frac{1}{32}(10 - 15 \cos 2\theta + 6 \cos 4\theta - \cos 6\theta) \quad [5]$$

- (b) Hence, using the substitution $x = \tan \theta$, find the exact value of

$$\int_0^1 \frac{x^6}{(1+x^2)^4} dx \quad [4]$$

Solution

- (a) Since

$$z = \cos \theta + i \sin \theta = e^{i\theta}$$

we have

$$z^{-1} = e^{-i\theta} = \cos \theta - i \sin \theta$$

and so

$$z - z^{-1} = e^{i\theta} - e^{-i\theta} = 2i \sin \theta$$

Hence

$$(z - z^{-1})^6 = (2i \sin \theta)^6 = 64i^6 \sin^6 \theta = -64 \sin^6 \theta$$

Now expand $(z - z^{-1})^6$ by the binomial theorem:

$$\begin{aligned} (z - z^{-1})^6 &= z^6 - 6z^5(z^{-1}) + 15z^4(z^{-1})^2 - 20z^3(z^{-1})^3 \\ &\quad + 15z^2(z^{-1})^4 - 6z(z^{-1})^5 + (z^{-1})^6 \\ &= z^6 - 6z^4 + 15z^2 - 20 + 15z^{-2} - 6z^{-4} + z^{-6} \end{aligned}$$

By de Moivre's theorem,

$$z^n = \cos n\theta + i \sin n\theta, \quad z^{-n} = \cos n\theta - i \sin n\theta$$

so

$$z^n + z^{-n} = 2 \cos n\theta$$

Therefore

$$\begin{aligned} (z - z^{-1})^6 &= (z^6 + z^{-6}) - 6(z^4 + z^{-4}) + 15(z^2 + z^{-2}) - 20 \\ &= 2 \cos 6\theta - 12 \cos 4\theta + 30 \cos 2\theta - 20 \end{aligned}$$

Equating this with the earlier result gives

$$-64 \sin^6 \theta = 2 \cos 6\theta - 12 \cos 4\theta + 30 \cos 2\theta - 20$$

so

$$64 \sin^6 \theta = 20 - 30 \cos 2\theta + 12 \cos 4\theta - 2 \cos 6\theta$$

and hence

$$\sin^6 \theta = \frac{1}{32}(10 - 15 \cos 2\theta + 6 \cos 4\theta - \cos 6\theta)$$

This is the required result.

- (b) Let

$$I = \int_0^1 \frac{x^6}{(1+x^2)^4} dx$$

Use the substitution

$$x = \tan \theta$$

Then

$$\frac{dx}{d\theta} = \sec^2 \theta \quad \Rightarrow \quad dx = \sec^2 \theta d\theta$$

and

$$1 + x^2 = 1 + \tan^2 \theta = \sec^2 \theta$$

Also, when $x = 0$, $\theta = 0$, and when $x = 1$, $\theta = \frac{\pi}{4}$.

So

$$\begin{aligned} I &= \int_0^{\pi/4} \frac{\tan^6 \theta}{(\sec^2 \theta)^4} \sec^2 \theta d\theta \\ &= \int_0^{\pi/4} \tan^6 \theta \sec^{-6} \theta d\theta \\ &= \int_0^{\pi/4} \frac{\sin^6 \theta}{\cos^6 \theta} \cos^6 \theta d\theta \\ &= \int_0^{\pi/4} \sin^6 \theta d\theta \end{aligned}$$

Using the result from part (a),

$$\sin^6 \theta = \frac{1}{32} (10 - 15 \cos 2\theta + 6 \cos 4\theta - \cos 6\theta)$$

Therefore

$$\begin{aligned} I &= \frac{1}{32} \int_0^{\pi/4} (10 - 15 \cos 2\theta + 6 \cos 4\theta - \cos 6\theta) d\theta \\ &= \frac{1}{32} \left[10\theta - \frac{15}{2} \sin 2\theta + \frac{3}{2} \sin 4\theta - \frac{1}{6} \sin 6\theta \right]_0^{\pi/4} \end{aligned}$$

Now substitute the limits:

$$\sin \frac{\pi}{2} = 1, \quad \sin \pi = 0, \quad \sin \frac{3\pi}{2} = -1$$

so

$$\begin{aligned} I &= \frac{1}{32} \left(10 \cdot \frac{\pi}{4} - \frac{15}{2} (1) + \frac{3}{2} (0) - \frac{1}{6} (-1) \right) \\ &= \frac{1}{32} \left(\frac{5\pi}{2} - \frac{15}{2} + \frac{1}{6} \right) \\ &= \frac{1}{32} \left(\frac{15\pi - 44}{6} \right) \\ &= \frac{15\pi - 44}{192} \end{aligned}$$

Hence the exact value is

$$\int_0^1 \frac{x^6}{(1+x^2)^4} dx = \frac{15\pi - 44}{192}$$

9. (a) Use de Moivre's theorem to show that

$$2 \cos 5\theta = 32 \cos^5 \theta - 40 \cos^3 \theta + 10 \cos \theta \quad [4]$$

(b) Hence solve the equation

$$x^5 - 5x^3 + 5x + 1 = 0$$

giving all roots in the form $2 \cos q\pi$, where q is a rational number. [4]

Solution

(a) By de Moivre's theorem,

$$(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$$

Expand the left-hand side using the binomial theorem:

$$\begin{aligned} (\cos \theta + i \sin \theta)^5 &= \cos^5 \theta + 5i \cos^4 \theta \sin \theta + 10i^2 \cos^3 \theta \sin^2 \theta \\ &\quad + 10i^3 \cos^2 \theta \sin^3 \theta + 5i^4 \cos \theta \sin^4 \theta + i^5 \sin^5 \theta \\ &= \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta \\ &\quad - 10i \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta \end{aligned}$$

Comparing real parts gives

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

Now use

$$\sin^2 \theta = 1 - \cos^2 \theta \quad \text{and} \quad \sin^4 \theta = (1 - \cos^2 \theta)^2$$

So

$$\begin{aligned} \cos 5\theta &= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - \cos^2 \theta)^2 \\ &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta (1 - 2 \cos^2 \theta + \cos^4 \theta) \\ &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta - 10 \cos^3 \theta + 5 \cos^5 \theta \\ &= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta \end{aligned}$$

Multiplying by 2,

$$2 \cos 5\theta = 32 \cos^5 \theta - 40 \cos^3 \theta + 10 \cos \theta$$

(b) Let

$$x = 2 \cos \theta$$

Then, using part (a),

$$\begin{aligned} x^5 - 5x^3 + 5x &= (2 \cos \theta)^5 - 5(2 \cos \theta)^3 + 5(2 \cos \theta) \\ &= 32 \cos^5 \theta - 40 \cos^3 \theta + 10 \cos \theta \\ &= 2 \cos 5\theta \end{aligned}$$

So the equation

$$x^5 - 5x^3 + 5x + 1 = 0$$

becomes

$$2 \cos 5\theta + 1 = 0$$

that is,

$$\cos 5\theta = -\frac{1}{2}$$

To obtain distinct values of $x = 2 \cos \theta$, take $0 \leq \theta \leq \pi$. Then $0 \leq 5\theta \leq 5\pi$, and in this interval

$$\cos 5\theta = -\frac{1}{2}$$

when

$$5\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3}, \frac{14\pi}{3}$$

Hence

$$\theta = \frac{2\pi}{15}, \frac{4\pi}{15}, \frac{8\pi}{15}, \frac{2\pi}{3}, \frac{14\pi}{15}$$

Therefore the roots are

$$x = 2 \cos \frac{2\pi}{15}, \quad 2 \cos \frac{4\pi}{15}, \quad 2 \cos \frac{8\pi}{15}, \quad 2 \cos \frac{2\pi}{3}, \quad 2 \cos \frac{14\pi}{15}$$

These are five distinct roots, so they are all the roots of the equation.

10. (a) If n is a positive integer and $z = \cos \theta + i \sin \theta$, show that

$$z^n + \frac{1}{z^n} = 2 \cos n\theta, \quad z^n - \frac{1}{z^n} = 2i \sin n\theta \quad [2]$$

(b) By considering $\left(z + \frac{1}{z}\right)^3 \left(z - \frac{1}{z}\right)^2$, find constants A , B and C such that

$$\sin^2 \theta \cos^3 \theta = A \cos 5\theta + B \cos 3\theta + C \cos \theta \quad [6]$$

Solution

(a) Write

$$z = \cos \theta + i \sin \theta = e^{i\theta}$$

Then by de Moivre's theorem,

$$z^n = e^{in\theta} = \cos n\theta + i \sin n\theta$$

Also,

$$\frac{1}{z} = e^{-i\theta} \quad \text{so} \quad \frac{1}{z^n} = z^{-n} = e^{-in\theta} = \cos n\theta - i \sin n\theta$$

Adding gives

$$\begin{aligned} z^n + \frac{1}{z^n} &= (\cos n\theta + i \sin n\theta) + (\cos n\theta - i \sin n\theta) \\ &= 2 \cos n\theta \end{aligned}$$

Subtracting gives

$$\begin{aligned} z^n - \frac{1}{z^n} &= (\cos n\theta + i \sin n\theta) - (\cos n\theta - i \sin n\theta) \\ &= 2i \sin n\theta \end{aligned}$$

Hence

$$z^n + \frac{1}{z^n} = 2 \cos n\theta, \quad z^n - \frac{1}{z^n} = 2i \sin n\theta$$

(b) From part (a), with $n = 1$,

$$z + \frac{1}{z} = 2 \cos \theta, \quad z - \frac{1}{z} = 2i \sin \theta$$

So

$$\begin{aligned} \left(z + \frac{1}{z}\right)^3 \left(z - \frac{1}{z}\right)^2 &= (2 \cos \theta)^3 (2i \sin \theta)^2 \\ &= 8 \cos^3 \theta \cdot 4i^2 \sin^2 \theta \\ &= 8 \cos^3 \theta \cdot (-4) \sin^2 \theta \\ &= -32 \sin^2 \theta \cos^3 \theta \end{aligned}$$

Now expand the expression algebraically.

First,

$$\left(z + \frac{1}{z}\right)^3 = z^3 + 3z + \frac{3}{z} + \frac{1}{z^3}$$

and

$$\left(z - \frac{1}{z}\right)^2 = z^2 - 2 + \frac{1}{z^2}$$

Hence

$$\begin{aligned}\left(z + \frac{1}{z}\right)^3 \left(z - \frac{1}{z}\right)^2 &= \left(z^3 + 3z + \frac{3}{z} + \frac{1}{z^3}\right) \left(z^2 - 2 + \frac{1}{z^2}\right) \\ &= (z^5 - 2z^3 + z) + \left(3z^3 - 6z + \frac{3}{z}\right) \\ &\quad + \left(3z - \frac{6}{z} + \frac{3}{z^3}\right) + \left(\frac{1}{z} - \frac{2}{z^3} + \frac{1}{z^5}\right) \\ &= z^5 + z^3 - 2z - \frac{2}{z} + \frac{1}{z^3} + \frac{1}{z^5}\end{aligned}$$

Group the terms in pairs:

$$\left(z^5 + \frac{1}{z^5}\right) + \left(z^3 + \frac{1}{z^3}\right) - 2\left(z + \frac{1}{z}\right)$$

Using part (a) again,

$$z^5 + \frac{1}{z^5} = 2 \cos 5\theta, \quad z^3 + \frac{1}{z^3} = 2 \cos 3\theta, \quad z + \frac{1}{z} = 2 \cos \theta$$

So

$$\left(z + \frac{1}{z}\right)^3 \left(z - \frac{1}{z}\right)^2 = 2 \cos 5\theta + 2 \cos 3\theta - 4 \cos \theta$$

Equating the two expressions,

$$-32 \sin^2 \theta \cos^3 \theta = 2 \cos 5\theta + 2 \cos 3\theta - 4 \cos \theta$$

Therefore

$$\begin{aligned}\sin^2 \theta \cos^3 \theta &= -\frac{1}{32} (2 \cos 5\theta + 2 \cos 3\theta - 4 \cos \theta) \\ &= -\frac{1}{16} \cos 5\theta - \frac{1}{16} \cos 3\theta + \frac{1}{8} \cos \theta\end{aligned}$$

So the constants are

$$A = -\frac{1}{16}, \quad B = -\frac{1}{16}, \quad C = \frac{1}{8}$$

11. (a) By writing $1 + i \tan \theta = \sec \theta (\cos \theta + i \sin \theta)$ and using de Moivre's theorem, show that

$$(1 + i \tan \theta)^4 = \frac{\cos 4\theta + i \sin 4\theta}{\cos^4 \theta} \quad [3]$$

- (b) Expand $(1 + i \tan \theta)^4$ and hence show that

$$\cos 4\theta = \frac{1 - 6 \tan^2 \theta + \tan^4 \theta}{(1 + \tan^2 \theta)^2}$$

and

$$\sin 4\theta = \frac{4 \tan \theta (1 - \tan^2 \theta)}{(1 + \tan^2 \theta)^2} \quad [4]$$

- (c) Deduce that

$$\tan 4\theta = \frac{4 \tan \theta (1 - \tan^2 \theta)}{1 - 6 \tan^2 \theta + \tan^4 \theta} \quad [2]$$

Solution

- (a) Starting with

$$1 + i \tan \theta = 1 + i \frac{\sin \theta}{\cos \theta} = \frac{\cos \theta + i \sin \theta}{\cos \theta} = \sec \theta (\cos \theta + i \sin \theta)$$

Raise both sides to the 4th power:

$$(1 + i \tan \theta)^4 = \sec^4 \theta (\cos \theta + i \sin \theta)^4$$

By de Moivre's theorem,

$$(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$$

so

$$(1 + i \tan \theta)^4 = \sec^4 \theta (\cos 4\theta + i \sin 4\theta) = \frac{\cos 4\theta + i \sin 4\theta}{\cos^4 \theta}$$

- (b) Let $t = \tan \theta$. Then

$$(1 + i \tan \theta)^4 = (1 + it)^4$$

Expanding:

$$\begin{aligned} (1 + it)^4 &= 1 + 4it + 6(it)^2 + 4(it)^3 + (it)^4 \\ &= 1 + 4it + 6i^2t^2 + 4i^3t^3 + i^4t^4 \\ &= 1 + 4it - 6t^2 - 4it^3 + t^4 \\ &= (1 - 6t^2 + t^4) + i(4t - 4t^3) \\ &= (1 - 6t^2 + t^4) + i4t(1 - t^2) \end{aligned}$$

From part (a),

$$(1 + it)^4 = \frac{\cos 4\theta + i \sin 4\theta}{\cos^4 \theta}$$

Comparing real and imaginary parts:

$$\frac{\cos 4\theta}{\cos^4 \theta} = 1 - 6t^2 + t^4 \quad \text{and} \quad \frac{\sin 4\theta}{\cos^4 \theta} = 4t(1 - t^2)$$

Now

$$1 + \tan^2 \theta = \sec^2 \theta = \frac{1}{\cos^2 \theta}$$

so

$$\cos^4 \theta = \frac{1}{(1 + \tan^2 \theta)^2} = \frac{1}{(1 + t^2)^2}$$

Therefore

$$\begin{aligned}\cos 4\theta &= (1 - 6t^2 + t^4) \cos^4 \theta \\ &= \frac{1 - 6t^2 + t^4}{(1 + t^2)^2}\end{aligned}$$

and

$$\begin{aligned}\sin 4\theta &= 4t(1 - t^2) \cos^4 \theta \\ &= \frac{4t(1 - t^2)}{(1 + t^2)^2}\end{aligned}$$

Replacing t by $\tan \theta$ gives

$$\cos 4\theta = \frac{1 - 6 \tan^2 \theta + \tan^4 \theta}{(1 + \tan^2 \theta)^2}$$

and

$$\sin 4\theta = \frac{4 \tan \theta (1 - \tan^2 \theta)}{(1 + \tan^2 \theta)^2}$$

(c) Using

$$\tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta}$$

and the results from part (b),

$$\begin{aligned}\tan 4\theta &= \frac{\frac{4 \tan \theta (1 - \tan^2 \theta)}{(1 + \tan^2 \theta)^2}}{\frac{1 - 6 \tan^2 \theta + \tan^4 \theta}{(1 + \tan^2 \theta)^2}} \\ &= \frac{4 \tan \theta (1 - \tan^2 \theta)}{1 - 6 \tan^2 \theta + \tan^4 \theta}\end{aligned}$$

12. (a) Use de Moivre's theorem to show that

$$\sec 6\theta = \frac{\sec^6 \theta}{32 - 48 \sec^2 \theta + 18 \sec^4 \theta - \sec^6 \theta} \quad [6]$$

(b) Hence obtain the roots of the equation

$$x^6 - 18x^4 + 48x^2 - 32 = 0$$

in the form $\sec(q\pi)$, where q is rational. [4]

Solution

(a) By de Moivre's theorem,

$$(\cos \theta + i \sin \theta)^6 = \cos 6\theta + i \sin 6\theta$$

Let $c = \cos \theta$ and $s = \sin \theta$. Expanding the left-hand side,

$$(c + is)^6 = c^6 + 6c^5(is) + 15c^4(is)^2 + 20c^3(is)^3 + 15c^2(is)^4 + 6c(is)^5 + (is)^6$$

Taking real parts gives

$$\cos 6\theta = c^6 - 15c^4s^2 + 15c^2s^4 - s^6$$

Now use $s^2 = 1 - c^2$:

$$\begin{aligned} \cos 6\theta &= c^6 - 15c^4(1 - c^2) + 15c^2(1 - c^2)^2 - (1 - c^2)^3 \\ &= c^6 - 15c^4 + 15c^6 + 15c^2(1 - 2c^2 + c^4) - (1 - 3c^2 + 3c^4 - c^6) \\ &= c^6 - 15c^4 + 15c^6 + 15c^2 - 30c^4 + 15c^6 - 1 + 3c^2 - 3c^4 + c^6 \\ &= 32c^6 - 48c^4 + 18c^2 - 1 \end{aligned}$$

So

$$\cos 6\theta = 32 \cos^6 \theta - 48 \cos^4 \theta + 18 \cos^2 \theta - 1$$

Hence

$$\sec 6\theta = \frac{1}{32 \cos^6 \theta - 48 \cos^4 \theta + 18 \cos^2 \theta - 1}$$

Multiply top and bottom by $\sec^6 \theta$:

$$\sec 6\theta = \frac{\sec^6 \theta}{32 - 48 \sec^2 \theta + 18 \sec^4 \theta - \sec^6 \theta}$$

(b) Let

$$x = \sec \theta$$

Then the equation

$$x^6 - 18x^4 + 48x^2 - 32 = 0$$

becomes

$$\sec^6 \theta - 18 \sec^4 \theta + 48 \sec^2 \theta - 32 = 0$$

or

$$32 - 48 \sec^2 \theta + 18 \sec^4 \theta - \sec^6 \theta = 0$$

From part (a),

$$\sec 6\theta = \frac{\sec^6 \theta}{32 - 48 \sec^2 \theta + 18 \sec^4 \theta - \sec^6 \theta}$$

So if

$$32 - 48 \sec^2 \theta + 18 \sec^4 \theta - \sec^6 \theta = 0$$

then $\sec 6\theta$ is undefined. Since $\sec^6 \theta \neq 0$, this means

$$\cos 6\theta = 0$$

Therefore

$$6\theta = \frac{(2k+1)\pi}{2}$$

so

$$\theta = \frac{(2k+1)\pi}{12}$$

Hence

$$x = \sec \left(\frac{(2k+1)\pi}{12} \right)$$

Taking $k = 0, 1, 2, 3, 4, 5$ gives the six distinct roots:

$$\sec \frac{\pi}{12}, \quad \sec \frac{\pi}{4}, \quad \sec \frac{5\pi}{12}, \quad \sec \frac{7\pi}{12}, \quad \sec \frac{3\pi}{4}, \quad \sec \frac{11\pi}{12}$$

So the roots are

$$x = \sec \left(\frac{(2k+1)\pi}{12} \right), \quad k = 0, 1, 2, 3, 4, 5$$

13. (a) Use de Moivre's theorem to show that

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta \quad [5]$$

(b) Hence solve for $0 \leq \theta \leq \pi$

$$16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta = \cos 2\theta$$

giving your answers as exact multiples of π .

[5]

Solution

(a) Let

$$z = \cos \theta + i \sin \theta$$

By de Moivre's theorem,

$$z^5 = \cos 5\theta + i \sin 5\theta$$

Now expand z^5 using the binomial theorem:

$$\begin{aligned} (\cos \theta + i \sin \theta)^5 &= \cos^5 \theta + 5 \cos^4 \theta (i \sin \theta) + 10 \cos^3 \theta (i \sin \theta)^2 \\ &\quad + 10 \cos^2 \theta (i \sin \theta)^3 + 5 \cos \theta (i \sin \theta)^4 + (i \sin \theta)^5 \\ &= \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta \\ &\quad - 10i \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta \end{aligned}$$

Equating real parts gives

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

Now use

$$\sin^2 \theta = 1 - \cos^2 \theta, \quad \sin^4 \theta = (1 - \cos^2 \theta)^2$$

So

$$\begin{aligned} \cos 5\theta &= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - \cos^2 \theta)^2 \\ &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta (1 - 2 \cos^2 \theta + \cos^4 \theta) \\ &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta - 10 \cos^3 \theta + 5 \cos^5 \theta \\ &= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta \end{aligned}$$

(b) Using part (a), the equation

$$16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta = \cos 2\theta$$

becomes

$$\cos 5\theta = \cos 2\theta$$

For $\cos A = \cos B$, the general solution is

$$A = 2n\pi \pm B \quad (n \in \mathbb{Z})$$

So

$$5\theta = 2n\pi \pm 2\theta$$

This gives two cases.

From

$$5\theta = 2n\pi + 2\theta$$

we get

$$3\theta = 2n\pi$$

so

$$\theta = \frac{2n\pi}{3}$$

For $0 \leq \theta \leq \pi$, this gives

$$\theta = 0, \frac{2\pi}{3}$$

From

$$5\theta = 2n\pi - 2\theta$$

we get

$$7\theta = 2n\pi$$

so

$$\theta = \frac{2n\pi}{7}$$

For $0 \leq \theta \leq \pi$, this gives

$$\theta = 0, \frac{2\pi}{7}, \frac{4\pi}{7}, \frac{6\pi}{7}$$

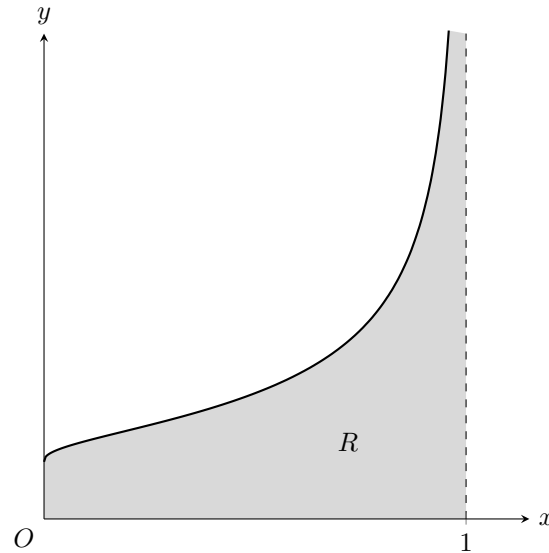
Combining the distinct solutions,

$$\theta = 0, \frac{2\pi}{7}, \frac{4\pi}{7}, \frac{2\pi}{3}, \frac{6\pi}{7}$$

These are the required exact values.

14. (a) Use de Moivre's theorem to determine constants A , B and C such that

$$\cos^5 \theta \equiv A \cos 5\theta + B \cos 3\theta + C \cos \theta \quad [5]$$



The function f is defined by

$$f(x) = -3 \sin \left(5 \cos^{-1} \left(x^{\frac{1}{6}} \right) \right) - 25 \sin \left(3 \cos^{-1} \left(x^{\frac{1}{6}} \right) \right) - 150 \sin \left(\cos^{-1} \left(x^{\frac{1}{6}} \right) \right), \quad 0 \leq x < 1$$

- (b) Show that

$$f'(x) = \frac{40}{\sqrt{1 - x^{\frac{1}{3}}}} \quad [6]$$

The diagram shows the curve with equation

$$y = \frac{1}{\sqrt{1 - x^{\frac{1}{3}}}}$$

for $0 \leq x < 1$ and the asymptote $x = 1$. The region R is the unbounded region between the curve, the x -axis, the line $x = 0$ and the line $x = 1$.

You are given that the area of R is finite.

- (c) Determine the exact area of R . [3]

Solution

- (a) Let $z = \cos \theta + i \sin \theta$. By de Moivre's theorem,

$$z^n + z^{-n} = (\cos n\theta + i \sin n\theta) + (\cos n\theta - i \sin n\theta) = 2 \cos n\theta$$

In particular, $z + z^{-1} = 2 \cos \theta$, so

$$(z + z^{-1})^5 = 32 \cos^5 \theta$$

Expanding by the binomial theorem,

$$\begin{aligned}(z + z^{-1})^5 &= z^5 + 5z^3 + 10z + 10z^{-1} + 5z^{-3} + z^{-5} \\ &= (z^5 + z^{-5}) + 5(z^3 + z^{-3}) + 10(z + z^{-1}) \\ &= 2 \cos 5\theta + 10 \cos 3\theta + 20 \cos \theta\end{aligned}$$

Equating the two expressions and dividing by 32,

$$\cos^5 \theta \equiv \frac{1}{16} \cos 5\theta + \frac{5}{16} \cos 3\theta + \frac{5}{8} \cos \theta$$

Hence

$$A = \frac{1}{16}, \quad B = \frac{5}{16}, \quad C = \frac{5}{8}$$

Alternative method (equating coefficients):

Let $c = \cos \theta$ and $s = \sin \theta$.

Using de Moivre's theorem,

$$(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$$

Expanding the left-hand side,

$$\begin{aligned}(c + is)^5 &= c^5 + 5c^4(is) + 10c^3(is)^2 + 10c^2(is)^3 + 5c(is)^4 + (is)^5 \\ &= c^5 + 5ic^4s - 10c^3s^2 - 10ic^2s^3 + 5cs^4 + is^5\end{aligned}$$

Taking real parts,

$$\cos 5\theta = c^5 - 10c^3s^2 + 5cs^4$$

Since $s^2 = 1 - c^2$,

$$\begin{aligned}\cos 5\theta &= c^5 - 10c^3(1 - c^2) + 5c(1 - c^2)^2 \\ &= c^5 - 10c^3 + 10c^5 + 5c(1 - 2c^2 + c^4) \\ &= 16c^5 - 20c^3 + 5c\end{aligned}$$

Similarly, from

$$(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$$

we get

$$\begin{aligned}(c + is)^3 &= c^3 + 3c^2(is) + 3c(is)^2 + (is)^3 \\ &= c^3 + 3ic^2s - 3cs^2 - is^3\end{aligned}$$

so taking real parts,

$$\begin{aligned}\cos 3\theta &= c^3 - 3cs^2 \\ &= c^3 - 3c(1 - c^2) \\ &= 4c^3 - 3c\end{aligned}$$

Now use

$$c^5 \equiv A \cos 5\theta + B \cos 3\theta + C \cos \theta$$

Substituting the expansions,

$$\begin{aligned}c^5 &\equiv A(16c^5 - 20c^3 + 5c) + B(4c^3 - 3c) + Cc \\ &\equiv 16Ac^5 + (-20A + 4B)c^3 + (5A - 3B + C)c\end{aligned}$$

Equating coefficients of powers of c ,

$$\begin{aligned}16A &= 1 \\ -20A + 4B &= 0 \\ 5A - 3B + C &= 0\end{aligned}$$

So

$$\begin{aligned} A &= \frac{1}{16} \\ -20\left(\frac{1}{16}\right) + 4B &= 0 \implies 4B = \frac{5}{4} \implies B = \frac{5}{16} \\ 5\left(\frac{1}{16}\right) - 3\left(\frac{5}{16}\right) + C &= 0 \implies -\frac{10}{16} + C = 0 \implies C = \frac{10}{16} = \frac{5}{8} \end{aligned}$$

Hence

$$A = \frac{1}{16}, \quad B = \frac{5}{16}, \quad C = \frac{5}{8}$$

(b) Let

$$\theta = \cos^{-1}\left(x^{1/6}\right)$$

so that

$$\cos \theta = x^{1/6}$$

Then

$$f(x) = -3 \sin 5\theta - 25 \sin 3\theta - 150 \sin \theta$$

Differentiate using the chain rule:

$$\begin{aligned} f'(x) &= -3(5 \cos 5\theta) \frac{d\theta}{dx} - 25(3 \cos 3\theta) \frac{d\theta}{dx} - 150(\cos \theta) \frac{d\theta}{dx} \\ &= -(15 \cos 5\theta + 75 \cos 3\theta + 150 \cos \theta) \frac{d\theta}{dx} \end{aligned}$$

Now

$$\theta = \cos^{-1}(x^{1/6})$$

and

$$\frac{d}{dx}(\cos^{-1} u) = -\frac{1}{\sqrt{1-u^2}} \frac{du}{dx}$$

Here $u = x^{1/6}$, so

$$\frac{du}{dx} = \frac{1}{6} x^{-5/6}$$

and therefore

$$\begin{aligned} \frac{d\theta}{dx} &= -\frac{1}{\sqrt{1-(x^{1/6})^2}} \cdot \frac{1}{6} x^{-5/6} \\ &= -\frac{x^{-5/6}}{6\sqrt{1-x^{1/3}}} \end{aligned}$$

Substitute this into the expression for $f'(x)$:

$$\begin{aligned} f'(x) &= -(15 \cos 5\theta + 75 \cos 3\theta + 150 \cos \theta) \left(-\frac{x^{-5/6}}{6\sqrt{1-x^{1/3}}} \right) \\ &= \frac{x^{-5/6}}{6\sqrt{1-x^{1/3}}} (15 \cos 5\theta + 75 \cos 3\theta + 150 \cos \theta) \\ &= \frac{5}{2} \cdot \frac{x^{-5/6}}{\sqrt{1-x^{1/3}}} (\cos 5\theta + 5 \cos 3\theta + 10 \cos \theta) \end{aligned}$$

From part (a),

$$\cos^5 \theta = \frac{1}{16} \cos 5\theta + \frac{5}{16} \cos 3\theta + \frac{5}{8} \cos \theta$$

Multiply by 16:

$$16 \cos^5 \theta = \cos 5\theta + 5 \cos 3\theta + 10 \cos \theta$$

So

$$f'(x) = \frac{5}{2} \cdot \frac{x^{-5/6}}{\sqrt{1-x^{1/3}}} \cdot 16 \cos^5 \theta$$

But $\cos \theta = x^{1/6}$, so

$$\cos^5 \theta = \left(x^{1/6}\right)^5 = x^{5/6}$$

Hence

$$\begin{aligned} f'(x) &= \frac{5}{2} \cdot \frac{x^{-5/6}}{\sqrt{1-x^{1/3}}} \cdot 16x^{5/6} \\ &= \frac{40}{\sqrt{1-x^{1/3}}} \end{aligned}$$

(c) The area of R is

$$\int_0^1 \frac{1}{\sqrt{1-x^{1/3}}} dx$$

Since the curve has asymptote $x = 1$, this is an improper integral:

$$\text{Area of } R = \lim_{b \rightarrow 1^-} \int_0^b \frac{1}{\sqrt{1-x^{1/3}}} dx$$

From part (b),

$$\frac{1}{\sqrt{1-x^{1/3}}} = \frac{1}{40} f'(x)$$

So

$$\begin{aligned} \text{Area of } R &= \frac{1}{40} \lim_{b \rightarrow 1^-} \int_0^b f'(x) dx \\ &= \frac{1}{40} \lim_{b \rightarrow 1^-} [f(x)]_0^b \\ &= \frac{1}{40} \left(\lim_{b \rightarrow 1^-} f(b) - f(0) \right) \end{aligned}$$

First find $\lim_{b \rightarrow 1^-} f(b)$.

As $b \rightarrow 1^-$,

$$b^{1/6} \rightarrow 1 \quad \text{so} \quad \cos^{-1}(b^{1/6}) \rightarrow 0$$

and therefore

$$f(b) \rightarrow -3 \sin 0 - 25 \sin 0 - 150 \sin 0 = 0$$

Now find $f(0)$:

$$\cos^{-1}(0^{1/6}) = \cos^{-1}(0) = \frac{\pi}{2}$$

so

$$\begin{aligned} f(0) &= -3 \sin \left(5 \cdot \frac{\pi}{2} \right) - 25 \sin \left(3 \cdot \frac{\pi}{2} \right) - 150 \sin \left(\frac{\pi}{2} \right) \\ &= -3(1) - 25(-1) - 150(1) \\ &= -3 + 25 - 150 \\ &= -128 \end{aligned}$$

Hence

$$\begin{aligned} \text{Area of } R &= \frac{1}{40} (0 - (-128)) \\ &= \frac{128}{40} \\ &= \frac{16}{5} \end{aligned}$$

The exact area of R is $\frac{16}{5}$.

15. (i) Use de Moivre's theorem to show that

$$\cos 5\theta \equiv \cos \theta (1 - 12 \sin^2 \theta + 16 \sin^4 \theta) \quad [5]$$

(ii) Hence determine the range of values of

$$\frac{\cos 5\theta}{\cos \theta}$$

given that $\cos \theta \neq 0$.

[7]

Solution

(i) Using de Moivre's theorem,

$$(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$$

Expand the left-hand side by the binomial theorem:

$$\begin{aligned} (\cos \theta + i \sin \theta)^5 &= \cos^5 \theta + 5 \cos^4 \theta (i \sin \theta) + 10 \cos^3 \theta (i \sin \theta)^2 \\ &\quad + 10 \cos^2 \theta (i \sin \theta)^3 + 5 \cos \theta (i \sin \theta)^4 + (i \sin \theta)^5 \end{aligned}$$

Now use

$$i^2 = -1, \quad i^3 = -i, \quad i^4 = 1, \quad i^5 = i$$

So

$$\begin{aligned} (\cos \theta + i \sin \theta)^5 &= \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta \\ &\quad - 10i \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta \end{aligned}$$

Equating real parts gives

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

Factor out $\cos \theta$:

$$\cos 5\theta = \cos \theta (\cos^4 \theta - 10 \cos^2 \theta \sin^2 \theta + 5 \sin^4 \theta)$$

Now use $\cos^2 \theta = 1 - \sin^2 \theta$:

$$\begin{aligned} \cos 5\theta &= \cos \theta ((1 - \sin^2 \theta)^2 - 10(1 - \sin^2 \theta) \sin^2 \theta + 5 \sin^4 \theta) \\ &= \cos \theta (1 - 2 \sin^2 \theta + \sin^4 \theta - 10 \sin^2 \theta + 10 \sin^4 \theta + 5 \sin^4 \theta) \\ &= \cos \theta (1 - 12 \sin^2 \theta + 16 \sin^4 \theta) \end{aligned}$$

Hence

$$\cos 5\theta \equiv \cos \theta (1 - 12 \sin^2 \theta + 16 \sin^4 \theta)$$

(ii) From part (i),

$$\frac{\cos 5\theta}{\cos \theta} = 1 - 12 \sin^2 \theta + 16 \sin^4 \theta$$

provided $\cos \theta \neq 0$.

Let

$$u = \sin^2 \theta$$

Then, since $0 \leq \sin^2 \theta \leq 1$ and $\cos \theta \neq 0$, we have

$$0 \leq u < 1$$

So we need the range of

$$f(u) = 1 - 12u + 16u^2 \quad \text{for } 0 \leq u < 1$$

This is a quadratic opening upwards, so its minimum occurs at the vertex:

$$u = \frac{-(-12)}{2 \cdot 16} = \frac{12}{32} = \frac{3}{8}$$

The minimum value is

$$\begin{aligned} f\left(\frac{3}{8}\right) &= 1 - 12\left(\frac{3}{8}\right) + 16\left(\frac{3}{8}\right)^2 \\ &= 1 - \frac{36}{8} + 16 \cdot \frac{9}{64} \\ &= 1 - \frac{9}{2} + \frac{9}{4} \\ &= \frac{4}{4} - \frac{18}{4} + \frac{9}{4} \\ &= -\frac{5}{4} \end{aligned}$$

So the least possible value is

$$-\frac{5}{4}$$

Now consider the largest possible value on $0 \leq u < 1$.

As $u \rightarrow 1^-$,

$$f(u) \rightarrow 1 - 12 + 16 = 5$$

We must check whether 5 is actually attained. Solve

$$\begin{aligned} 1 - 12u + 16u^2 &= 5 \\ 16u^2 - 12u - 4 &= 0 \\ 4u^2 - 3u - 1 &= 0 \\ (4u + 1)(u - 1) &= 0 \end{aligned}$$

So

$$u = 1 \quad \text{or} \quad u = -\frac{1}{4}$$

But $u = \sin^2 \theta$ cannot be negative, and $u = 1$ would mean

$$\sin^2 \theta = 1 \implies \cos \theta = 0$$

which is excluded.

So 5 is not attained.

Therefore the range of

$$\frac{\cos 5\theta}{\cos \theta}$$

is

$$-\frac{5}{4} \leq \frac{\cos 5\theta}{\cos \theta} < 5$$

16. (a) Use de Moivre's theorem to show that

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta \quad [5]$$

(b) Use the identity given in part (a) to find the 2 positive roots of

$$64x^5 - 80x^3 + 20x + 1 = 0$$

giving your answers to 3 significant figures. [4]

Solution

(a) Let $c = \cos \theta$ and $s = \sin \theta$. By de Moivre's theorem,

$$(c + is)^5 = \cos 5\theta + i \sin 5\theta$$

Expanding the left-hand side,

$$\begin{aligned} (c + is)^5 &= c^5 + 5c^4(is) + 10c^3(is)^2 + 10c^2(is)^3 + 5c(is)^4 + (is)^5 \\ &= c^5 + 5ic^4s - 10c^3s^2 - 10ic^2s^3 + 5cs^4 + is^5 \end{aligned}$$

Equating real parts gives

$$\cos 5\theta = c^5 - 10c^3s^2 + 5cs^4$$

Now use $s^2 = 1 - c^2$ and $s^4 = (1 - c^2)^2$:

$$\begin{aligned} \cos 5\theta &= c^5 - 10c^3(1 - c^2) + 5c(1 - c^2)^2 \\ &= c^5 - 10c^3 + 10c^5 + 5c(1 - 2c^2 + c^4) \\ &= c^5 - 10c^3 + 10c^5 + 5c - 10c^3 + 5c^5 \\ &= 16c^5 - 20c^3 + 5c \end{aligned}$$

Since $c = \cos \theta$,

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$$

as required.

(b) Let

$$x = \cos \theta \quad \text{with } 0^\circ \leq \theta \leq 180^\circ$$

Then

$$\begin{aligned} 64x^5 - 80x^3 + 20x + 1 &= 0 \\ 4(16x^5 - 20x^3 + 5x) + 1 &= 0 \end{aligned}$$

Using the identity from part (a),

$$16x^5 - 20x^3 + 5x = \cos 5\theta$$

so

$$4 \cos 5\theta + 1 = 0$$

Hence

$$\cos 5\theta = -\frac{1}{4}$$

Let

$$\alpha = \arccos\left(-\frac{1}{4}\right) \approx 104.478^\circ$$

Since $0^\circ \leq \theta \leq 180^\circ$, we have $0^\circ \leq 5\theta \leq 900^\circ$. Therefore the solutions for 5θ in this interval are

$$\alpha, \quad 360^\circ - \alpha, \quad 360^\circ + \alpha, \quad 720^\circ - \alpha, \quad 720^\circ + \alpha$$

So

$$\theta \approx 20.896^\circ, 51.104^\circ, 92.896^\circ, 123.104^\circ, 164.896^\circ$$

Now $x = \cos \theta$, and $x > 0$ when $0^\circ < \theta < 90^\circ$. So only the first two values give positive roots:

$$x = \cos 20.896^\circ \approx 0.934$$

$$x = \cos 51.104^\circ \approx 0.628$$

So the two positive roots are $x \approx 0.628$ and $x \approx 0.934$.

17. (a) Use de Moivre's theorem to show that

$$\sin^2 \theta \cos^2 \theta = \frac{1}{8}(1 - \cos 4\theta) \quad [5]$$

(b) Hence, or otherwise, find the solution of the differential equation

$$\frac{dy}{d\theta} - 2y \tan \theta = \sin^2 \theta$$

for which $y = 0$ when $\theta = \frac{1}{4}\pi$.

[6]

Solution

(a) By de Moivre's theorem,

$$(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$$

Expanding the left-hand side,

$$\begin{aligned} (\cos \theta + i \sin \theta)^4 &= \cos^4 \theta + 4 \cos^3 \theta (i \sin \theta) + 6 \cos^2 \theta (i \sin \theta)^2 \\ &\quad + 4 \cos \theta (i \sin \theta)^3 + (i \sin \theta)^4 \\ &= \cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta \\ &\quad - 4i \cos \theta \sin^3 \theta + \sin^4 \theta \end{aligned}$$

So, comparing real parts,

$$\cos 4\theta = \cos^4 \theta - 6 \sin^2 \theta \cos^2 \theta + \sin^4 \theta$$

Now rewrite the right-hand side as

$$\begin{aligned} \cos^4 \theta - 6 \sin^2 \theta \cos^2 \theta + \sin^4 \theta &= (\sin^2 \theta + \cos^2 \theta)^2 - 8 \sin^2 \theta \cos^2 \theta \\ &= 1 - 8 \sin^2 \theta \cos^2 \theta \end{aligned}$$

Hence

$$\cos 4\theta = 1 - 8 \sin^2 \theta \cos^2 \theta$$

Therefore,

$$\sin^2 \theta \cos^2 \theta = \frac{1 - \cos 4\theta}{8}$$

So

$$\sin^2 \theta \cos^2 \theta = \frac{1}{8}(1 - \cos 4\theta)$$

(b) The differential equation is

$$\frac{dy}{d\theta} - 2y \tan \theta = \sin^2 \theta$$

This is a linear first order differential equation of the form

$$\frac{dy}{d\theta} + P(\theta)y = Q(\theta)$$

with

$$P(\theta) = -2 \tan \theta$$

The integrating factor is

$$\mu(\theta) = e^{\int -2 \tan \theta \, d\theta}$$

Using

$$\int \tan \theta \, d\theta = -\ln |\cos \theta|$$

we get

$$\mu(\theta) = e^{2 \ln |\cos \theta|} = \cos^2 \theta$$

Multiply the differential equation through by $\cos^2 \theta$:

$$\cos^2 \theta \frac{dy}{d\theta} - 2y \sin \theta \cos \theta = \sin^2 \theta \cos^2 \theta$$

The left-hand side is

$$\frac{d}{d\theta}(y \cos^2 \theta)$$

So

$$\frac{d}{d\theta}(y \cos^2 \theta) = \sin^2 \theta \cos^2 \theta$$

Using part (a),

$$\frac{d}{d\theta}(y \cos^2 \theta) = \frac{1 - \cos 4\theta}{8}$$

Integrating,

$$\begin{aligned} y \cos^2 \theta &= \int \frac{1 - \cos 4\theta}{8} d\theta \\ &= \frac{1}{8} \int 1 d\theta - \frac{1}{8} \int \cos 4\theta d\theta \\ &= \frac{\theta}{8} - \frac{\sin 4\theta}{32} + C \end{aligned}$$

Now use the condition $y = 0$ when $\theta = \frac{\pi}{4}$:

$$0 \cdot \cos^2 \frac{\pi}{4} = \frac{1}{8} \cdot \frac{\pi}{4} - \frac{\sin \pi}{32} + C$$

So

$$0 = \frac{\pi}{32} + C$$

and hence

$$C = -\frac{\pi}{32}$$

Therefore,

$$y \cos^2 \theta = \frac{\theta}{8} - \frac{\sin 4\theta}{32} - \frac{\pi}{32}$$

So

$$y = \frac{\frac{4\theta}{32} - \frac{\sin 4\theta}{32} - \frac{\pi}{32}}{\cos^2 \theta} = \frac{4\theta - \sin 4\theta - \pi}{32 \cos^2 \theta}$$

18. (a) Use de Moivre's theorem to show that

$$\cos^6 \theta = a \cos 6\theta + b \cos 4\theta + c \cos 2\theta + d$$

where a , b , c and d are constants to be found.

[5]

(b) Hence show that

$$\int_0^{\pi/4} \cos^6 \theta \, d\theta = \frac{11}{48} + \frac{5\pi}{64}$$

[5]

Solution

(a) Let

$$z = e^{i\theta}$$

Then

$$\cos \theta = \frac{z + z^{-1}}{2}$$

and, by de Moivre's theorem,

$$z^n + z^{-n} = e^{in\theta} + e^{-in\theta} = 2 \cos n\theta$$

Now

$$64 \cos^6 \theta = (z + z^{-1})^6$$

Expanding by the binomial theorem,

$$(z + z^{-1})^6 = z^6 + 6z^4 + 15z^2 + 20 + 15z^{-2} + 6z^{-4} + z^{-6}$$

Group the terms in pairs:

$$\begin{aligned} 64 \cos^6 \theta &= (z^6 + z^{-6}) + 6(z^4 + z^{-4}) + 15(z^2 + z^{-2}) + 20 \\ &= 2 \cos 6\theta + 12 \cos 4\theta + 30 \cos 2\theta + 20 \end{aligned}$$

Dividing by 64,

$$\begin{aligned} \cos^6 \theta &= \frac{1}{32} \cos 6\theta + \frac{12}{64} \cos 4\theta + \frac{30}{64} \cos 2\theta + \frac{20}{64} \\ &= \frac{1}{32} \cos 6\theta + \frac{3}{16} \cos 4\theta + \frac{15}{32} \cos 2\theta + \frac{5}{16} \end{aligned}$$

Hence

$$a = \frac{1}{32}, \quad b = \frac{3}{16}, \quad c = \frac{15}{32}, \quad d = \frac{5}{16}$$

(b) Using the identity from part (a),

$$\cos^6 \theta = \frac{1}{32} (\cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 10)$$

So

$$\begin{aligned} \int_0^{\pi/4} \cos^6 \theta \, d\theta &= \frac{1}{32} \int_0^{\pi/4} (\cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 10) \, d\theta \\ &= \frac{1}{32} \left[\frac{1}{6} \sin 6\theta + \frac{6}{4} \sin 4\theta + \frac{15}{2} \sin 2\theta + 10\theta \right]_0^{\pi/4} \\ &= \frac{1}{32} \left[\frac{1}{6} \sin 6\theta + \frac{3}{2} \sin 4\theta + \frac{15}{2} \sin 2\theta + 10\theta \right]_0^{\pi/4} \end{aligned}$$

At $\theta = \frac{\pi}{4}$,

$$\sin \frac{3\pi}{2} = -1, \quad \sin \pi = 0, \quad \sin \frac{\pi}{2} = 1$$

Therefore

$$\begin{aligned} \int_0^{\pi/4} \cos^6 \theta \, d\theta &= \frac{1}{32} \left(-\frac{1}{6} + \frac{3}{2} \cdot 0 + \frac{15}{2} \cdot 1 + 10 \cdot \frac{\pi}{4} \right) \\ &= \frac{1}{32} \left(-\frac{1}{6} + \frac{15}{2} + \frac{5\pi}{2} \right) \\ &= \frac{1}{32} \left(\frac{44}{6} + \frac{5\pi}{2} \right) \\ &= \frac{11}{48} + \frac{5\pi}{64} \end{aligned}$$